

fail2ban Cheat Sheet

Watch logs for repeated failed logins and block the guilty IP addresses in the firewall for a while. Commands need sudo.

How it works

```
sshd ---> log or journal ---> filter (regex) ---> 5 failures in 10 min? ---> firewall ban for 1 h
                                   picks out the IP      maxretry / findtime      bantime, then unban
```

A **jail** = one log + one filter + one action, e.g. **sshd** or **nginx-http-auth**. Only enabled jails run.

Using it

```
fail2ban-client status  ← list the running jails
fail2ban-client status sshd  ← counts and banned IPs for one jail
fail2ban-client set sshd unbanip 203.0.113.45 ← let one IP back in to this jail
fail2ban-client set sshd banip 203.0.113.99 ← ban an IP by hand
fail2ban-client unban 203.0.113.45      fail2ban-client unban --all ← unban from every jail · unban everyone
fail2ban-client get sshd bantime  ← read a jail setting (findtime, maxretry, ignoreip...)
fail2ban-client reload  ← re-read the config; current bans stay
fail2ban-client banned  ← every banned IP, per jail (v1.0+)
```

Reading the status output

```
$ fail2ban-client status
Status
|- Number of jail:      2
`- Jail list:  nginx-http-auth, sshd  ← jail names to use in the other commands

$ fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 3  ← IPs with failures counted, not banned yet
| |- Total failed:     1287 ← failed attempts seen since fail2ban started
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd  ← reads the journal (or File list: /var/log/auth.log)
`- Actions
   |- Currently banned: 2  ← blocked right now
   |- Total banned:     94  ← bans since start (counts reset on restart)
   `-- Banned IP list:   203.0.113.45 203.0.113.77 ← who is blocked
```

Configuring a jail

Never edit **jail.conf** (updates overwrite it). Put changes in **/etc/fail2ban/jail.local** or **/etc/fail2ban/jail.d/*.local**, then **fail2ban-client reload**.

```
[DEFAULT]  ← applies to every jail
bantime   = 1h  ← how long a ban lasts (-1 = forever)
findtime  = 10m ← window for counting failures
maxretry  = 5   ← failures within findtime that trigger a ban
ignoreip  = 127.0.0.1/8 ::1 192.168.1.0/24 ← never ban these: add your own IP or network
bantime.increment = true  ← repeat offenders get longer bans each time

[sshd]
enabled = true
backend = systemd  ← when there is no /var/log/auth.log (Debian) or /var/log/secure (RHEL)
```

Testing and troubleshooting

```
fail2ban-regex /var/log/auth.log /etc/fail2ban/filter.d/sshd.conf ← test a filter: want Lines: ... matched > 0
fail2ban-regex systemd-journal sshd ← same test against the journal
fail2ban-client -t ← check the config for errors before reloading
tail -f /var/log/fail2ban.log ← live Found / Ban / Unban lines and errors
nft list ruleset | grep -A5 f2b ← the bans in the firewall (nftables action)
iptables -L f2b-sshd -n ← same, for the older iptables actions
```

Total failed stuck at 0 while attacks happen = wrong log or a filter that doesn't match. **fail2ban-regex** shows which.

Remember: add your own IP to **ignoreip** before you lock yourself out. If you do get banned, unban from a console or another IP with **set sshd unbanip**.